

AI Engineer Take-Home Assignment

Objective

Build a To-Do List & Calendar Web Application that allows users to organize their daily tasks through an interactive calendar interface. The application should support authentication, task management, and task status tracking.

Functional Requirements

1. User Authentication

- User Registration
- User Login
- User Logout
- Protected routes/pages
- Each user should only be able to access and manage their own tasks.

2. Calendar

- View tasks by date
- Select any date on the calendar
- Display all tasks associated with the selected date
- Create new tasks directly for the selected date

3. Task Management

Users should be able to:

- Create a task
- Edit a task
- Delete a task
- View task details

Each task should include:

- Title
- Description
- Date
- Status

Available task statuses:

- Not Started
- In Progress
- Done

4. Dashboard

- Calendar view
- Task list for the selected date
- Summary of tasks grouped by status (Not Started, In Progress, Done)

5. Database

Persist all user and task data in a database. Local storage alone is not sufficient.

Technical Requirements

Candidates are free to choose their preferred technology stack.

Frontend

- React
- Next.js
- Vue.js
- Angular

Backend

- Node.js (Express/NestJS)
- Next.js API Routes
- Laravel
- Django
- Go
- ASP.NET

Database

- PostgreSQL
- MySQL
- MongoDB
- SQLite

Authentication

- JWT
- Session-based Authentication
- NextAuth/Auth.js
- Clerk
- Firebase Auth
- Any secure authentication solution

Bonus Points

- Task filtering by status
- Search functionality
- Responsive UI
- Modern UI/UX
- Drag-and-drop task management
- Dark mode
- Email verification
- API documentation
- Unit or integration tests
- Docker support
- CI/CD setup
- Deployment
- Well-structured Git commit history

Evaluation Criteria

- Completeness of required features
- Code quality and project structure
- Software architecture
- Database design
- UI/UX quality

- Authentication and security
- Performance and maintainability
- Error handling
- Documentation quality
- Overall engineering best practices

Submission Instructions

- Submit a public GitHub repository.
- Include a comprehensive README (project overview, installation, environment setup, how to run, technology stack).
- Provide demo credentials if applicable.
- Include a deployment link (optional but recommended).
- Email the public GitHub repository link to admin@aidece.ai.

Deadline

June 30, 2026 — 12:00 PM (WIB / GMT+7)

Late submissions may not be considered.